

EMMANUEL NWORIE

PhD Candidate, Computational and Applied Mathematics

Dallas, TX | enworie@smu.edu | 214-606-4184

PROFESSIONAL PROFILE

Mathematics educator and researcher with university-level teaching experience in calculus, linear algebra, differential equations, statistics, and college algebra. Committed to clear instruction, active learning, and helping students build confidence in quantitative reasoning. Research interests include mathematical modeling, partial differential equations, and scientific computing.

EDUCATION

PhD, Computational and Applied Mathematics, Southern Methodist University, 2021 - Present

MS, Computational and Applied Mathematics, Southern Methodist University, 2023

BS, Mathematics (Honors), University of Nigeria, Nsukka, 2015 - 2018

TEACHING EXPERIENCE

Graduate Tutor | Altshuler Learning Enhancement Center, Southern Methodist University *Aug 2024 - Present*
Dallas, Texas

- Provide individualized and small-group support in Calculus I, II, and III, Linear Algebra, Differential Equations, Analysis, College Algebra, Mathematical Modeling, and Introductory Statistics.
- Develop structured learning plans tailored to individual preparation levels and learning styles.
- Diagnose conceptual misunderstandings and design targeted interventions that improve student performance and confidence.

Graduate Teaching Assistant | Southern Methodist University
Dallas, Texas

Aug 2021 - Aug 2024

- Supported instruction in core undergraduate mathematics courses including Calculus I, II, and III, Linear Algebra, Differential Equations, and Mathematical Biology.
- Led recitation sessions, held office hours, and facilitated collaborative problem solving workshops.
- Prepared review materials, graded assignments, and provided detailed feedback aligned with course outcomes.

Advanced Level Mathematics Instructor | Doyen Academy
Enugu, Nigeria

Jan 2019 - Jul 2021

- Taught advanced secondary mathematics including algebra, trigonometry, coordinate geometry, statistics, elementary mechanics, and applied mathematics.
- Designed curriculum and assessments aligned with national examination standards.
- Prepared students for university entrance examinations and Advanced Level examinations.

Mathematics Tutor | Our Lady of Lourdes Secondary School
Ebonyi, Nigeria

Aug 2008 - Jun 2012

- Provided foundational mathematics instruction through individualized and group tutoring.

RESEARCH

Diffusion-Limited Growth Modeling in Biological Systems | Southern Methodist University

- Develop finite-element and PDE models for transport phenomena in biological systems.
- Implement numerical solvers and computational experiments in Python and related scientific computing tools.
- Collaborate across disciplines and present research findings in academic settings.

TEACHING AREAS

Calculus I-III, Linear Algebra, Differential Equations, Analysis, College Algebra, Introductory Statistics, Mathematical Biology, Mathematical Modeling, Advanced Secondary Mathematics

TECHNICAL SKILLS

Programming: Python, MATLAB, R

Scientific Computing: NumPy, SciPy, pandas, Matplotlib, Jupyter

Methods: Finite elements, PDE modeling, numerical methods, sparse solvers

Tools: LaTeX, Linux, Git, visualization software

SELECTED PROFESSIONAL DEVELOPMENT

Introduction to Artificial Intelligence with Python, Harvard Online
Machine Learning with Python, Stanford University Online
Introduction to Data Science with Python, Harvard Online
R for Data Analytics, Southern Methodist University

AWARDS AND HONORS

Betty McKnight Spears Graduate Student Teaching Award in Mathematics
R. Haberman Fellowship
H.E. Emerole Award in Mathematics
National Mathematics Competition - Gold and Silver Medals